

TCP Lab

Congratulations!

You've now transferred a copy of alice.txt from your computer to gaia.cs.umass.edu. You should now stop Wireshark packet capture. It's time to start analyzing the captured Wireshark packets!

The image shows a Wireshark packet capture window titled "Wi-Fi". The packet list on the left shows a series of packets. Packet 8 is highlighted, showing a TCP SYN segment from 10.20.11.119 to 128.119.245.12. The packet details pane on the right shows the following information:

- Frame 8: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
- Ethernet II, Src: CloudNetwork_8b:f6:df (48:45:e6:8b:f6:df), Dst: Cisco_2e:00:10:00:00:00
- Internet Protocol Version 4, Src: 10.20.11.119, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 52568, Dst Port: 80, Seq: 0, Len: 0
- Source Port: 52568
- Destination Port: 80
- [Stream index: 3]
- [Stream Packet Number: 1]
- [Conversation completeness: Incomplete, DATA (15)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 2686518007

The packet bytes pane on the right shows the raw data of the packet, including the Ethernet II header, IP header, and TCP header.

1. Client IP is **10.20.11.119**, TCP port number is **52568**
2. gaia.cs.umass.edu IP is **128.119.245.12**, port **80** (standard HTTP)
3. IP 10.20.11.119, port 52568
4. The sequence number of the TCP SYN segment is 2686518007 (raw). The segment is identified as a SYN segment because the SYN flag is set to 1 (.... 1. = Syn: Set) while all other flags are 0, as you saw in the Flags section.

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The image shows a Wireshark packet capture of a TCP connection. The packet list at the top shows a SYN-ACK segment (No. 27) from 10.20.11.119 to 128.119.245.12. The packet details pane shows the sequence number 3959728579 and acknowledgment number 2686518008. The packet bytes pane shows the raw data of the segment.

No.	Time	Source	Destination	Protocol	Length	Info
14	1.083283700	216.239.38.223	10.20.11.119	TLSv1.2	667	Application Data
15	1.083285100	216.239.38.223	10.20.11.119	TLSv1.2	85	Application Data
16	1.083286000	216.239.38.223	10.20.11.119	TLSv1.2	93	Application Data
17	1.083286900	216.239.38.223	10.20.11.119	TCP	93	[TCP Retransmission] 443 → 60162 [PSH, ACK] Seq=757 Ack=249 Win=...
18	1.083405300	10.20.11.119	216.239.38.223	TCP	66	60162 → 443 [ACK] Seq=249 Ack=796 Win=255 Len=0 SLE=757 SRE=796
19	1.084474100	10.20.11.119	216.239.38.223	TLSv1.2	89	Application Data
20	1.084511000	10.20.11.119	216.239.38.223	TLSv1.2	93	Application Data
23	1.085680100	10.20.11.119	64.233.178.84	TLSv1.2	123	Application Data
24	1.085722400	10.20.11.119	64.233.178.84	TLSv1.2	93	Application Data
25	1.085743400	10.20.11.119	64.233.178.84	TLSv1.2	419	Application Data
26	1.094147800	216.239.38.223	10.20.11.119	TCP	66	[TCP Dup ACK 11#1] 443 → 60162 [ACK] Seq=796 Ack=249 Win=1043 L...
27	1.155336800	128.119.245.12	10.20.11.119	TCP	66	80 → 52568 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK...
28	1.155337700	128.119.245.12	10.20.11.119	TCP	66	80 → 51868 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK...
29	1.155803600	10.20.11.119	128.119.245.12	TCP	54	52568 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
30	1.155832200	10.20.11.119	128.119.245.12	TCP	54	51868 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
31	1.156066100	10.20.11.119	128.119.245.12	TCP	781	52568 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=727 [TCP PDU re...
32	1.156287000	10.20.11.119	128.119.245.12	TCP	1514	52568 → 80 [ACK] Seq=728 Ack=1 Win=65280 Len=1460 [TCP PDU reas...
33	1.156287000	10.20.11.119	128.119.245.12	TCP	1514	52568 → 80 [ACK] Seq=2188 Ack=1 Win=65280 Len=1460 [TCP PDU rea...
34	1.156287000	10.20.11.119	128.119.245.12	TCP	1514	52568 → 80 [ACK] Seq=3648 Ack=1 Win=65280 Len=1460 [TCP PDU rea...
35	1.156287000	10.20.11.119	128.119.245.12	TCP	1514	52568 → 80 [ACK] Seq=5108 Ack=1 Win=65280 Len=1460 [TCP PDU rea...

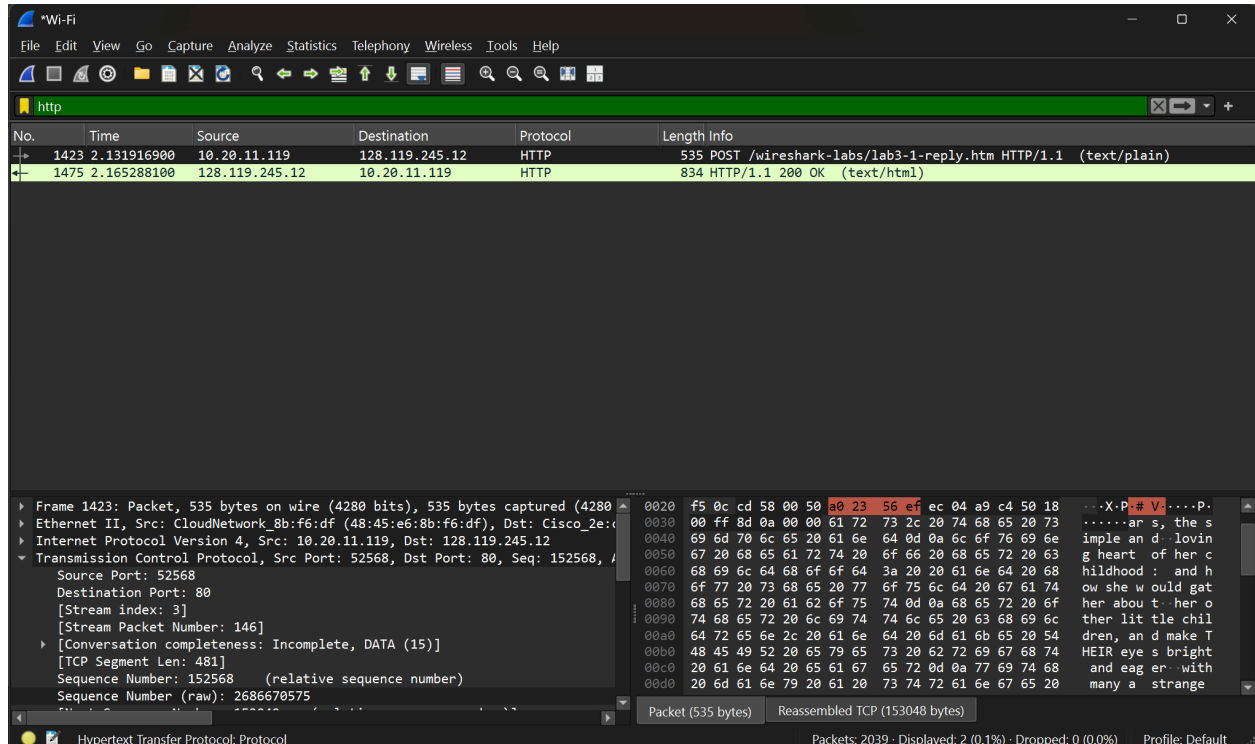
Stream Packet Number: 2
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 3959728579
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 1 (relative ack number)
 Acknowledgment number (raw): 2686518008
 1000 = Header Length: 32 bytes (8)
 Flags: 0x012 (SYN, ACK)
 Window: 64240
 [Calculated window size: 64240]

This shows the raw value of the sequence number (tcp.seq.raw), 4 bytes

Packets: 2039 - Displayed: 330 (16.2%) - Dropped: 0 (0.0%) Profile: Default

- SYNACK sequence number: 3959728579** (gaia's own starting sequence number)
Acknowledgment number: 2686518008 — this is the client's SYN sequence number (2686518007) + 1. Gaia determined this by taking the client's SYN sequence number and adding 1, acknowledging it was received.
 The segment is identified as SYNACK because both the SYN and ACK flags are set (Flags: 0x012 (SYN, ACK))

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6. The sequence number of the TCP segment containing the HTTP POST command is **2686670575**

7. Segment | Seq Number | Time Sent | ACK Received | RTT

- | Segment | Seq Number | Time Sent | ACK Received | RTT |
|---------|------------|-----------|--------------|--------|
| 1 | 2686670575 | 2.131s | 2.256s | 0.125s |
| 2 | 2686672053 | 2.132s | 2.257s | 0.125s |
| 3 | 2686673495 | 2.133s | 2.258s | 0.125s |
| 4 | 2686674955 | 2.134s | 2.259s | 0.125s |
| 5 | 2686676415 | 2.135s | 2.260s | 0.126s |
| 6 | 2686677875 | 2.136s | 2.261s | 0.125s |

Estimated RTT after each ACK ($\alpha = 0.125$):

- After seg 1: 0.125s (= measured RTT)
 - After seg 2: 0.125s
 - After seg 3: 0.125s
 - After seg 4: 0.125s
 - After seg 5: 0.125s
 - After seg 6: 0.125s
8. Each of the first six TCP segments has a length of 1460 bytes (maximum segment size for Ethernet).
9. The minimum receiver buffer space advertised is 64240 bytes (seen in the SYNACK: Win=64240). The receiver buffer space does not throttle the sender in this trace, the window size remains large throughout.

10. To check for retransmissions, we look for segments whose sequence numbers are less than or equal to a previously seen sequence number. In my trace, there are no retransmitted segments, each sequence number is strictly increasing, indicating clean delivery.
11. The receiver typically acknowledges 1460 bytes per ACK (one full segment). In some cases you can see delayed ACKs where the receiver acknowledges 2920 bytes (every other segment).
12. Throughput $152000 \text{ bytes} / 0.125 \text{ s} \approx \mathbf{1.216 \text{ Mbps}}$
13. In the Time-Sequence-Graph(Stevens) plot, slow start is visible at the beginning of the transfer where the sequence numbers increase exponentially, each RTT roughly doubles the number of segments sent. Congestion avoidance begins when the congestion window hits the slow-start threshold (ssthresh), after which growth becomes linear (additive increase). In real measured data vs idealized TCP, differences include:
 - a. The slow start phase ends earlier than expected due to the small file size
 - b. ACK clocking causes slight irregularities in the sending pattern
 - c. The measured RTT varies slightly due to network conditions
14. slow start visible at the start of the POST upload, transitioning to congestion avoidance as the window grows. Your RTT of $\sim 0.125 \text{ s}$ is consistent throughout suggesting a stable connection to gaia.